

PC-05 · UV FILTERS · PERSONAL CARE

HS MBBT

Methylene Bis-Benzotriazolyl Tetramethylbutylphenol

INCI	CAS	EC	FAMILY	SUPPLIED AS
Methylene Bis-Benzotriazolyl Tetramethylbutylphenol	103597-45-1	403-800-1	PC-05	Per filter

01 POSITIONING

SPF economics are filter economics — broad-spectrum molecules do more of the work with less of the load.

02 HOW IT WORKS

- 01

Dual-band absorption

Extended chromophores absorb across 280–400 nm, flattening the protection curve.
- 02

Photostability

Modern broad-spectrum classes hold absorbance through solar exposure — no SPF decay, and they photostabilize fragile partners.
- 03

Load efficiency

More protection per gram means lighter textures at equal SPF.

03 APPLICATIONS

- High-SPF face and body sunscreens
- Daily-wear SPF moisturizers
- Anti-aging lines with UVA claims
- Color cosmetics with SPF

04 FORMULATION GUIDE

USE LEVEL	Per regional annexes
PH & CONDITIONS	Formulate per regional positive lists (EU Annex VI, MERCOSUR, FDA); respect ceilings per filter

- Dissolve crystalline filters hot in ester solvents (IPM/IPP class) before emulsification.
- Combine with UVB boosters to reach target SPF at minimal total load.

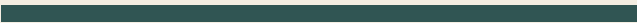
05 TYPICAL SPECIFICATION

Appearance	per filter: powder / viscous liquid
Assay	≥ 98%
Absorption range	280–400 nm
Regulatory listing	per region, per TDS

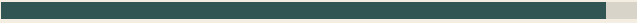
Typical parameters. The specification on the current TDS and the per-lot Certificate of Analysis govern.

06 PERFORMANCE – DIRECTIONAL

Protection breadth (UVA+UVB)
Broad-spectrum class



Photostability after 30 MED (class data)
Stabilized broad-spectrum systems



Values are representative of the ingredient class, based on published technical literature, and shown directionally for comparison. Final performance must be validated in the customer's formulation.